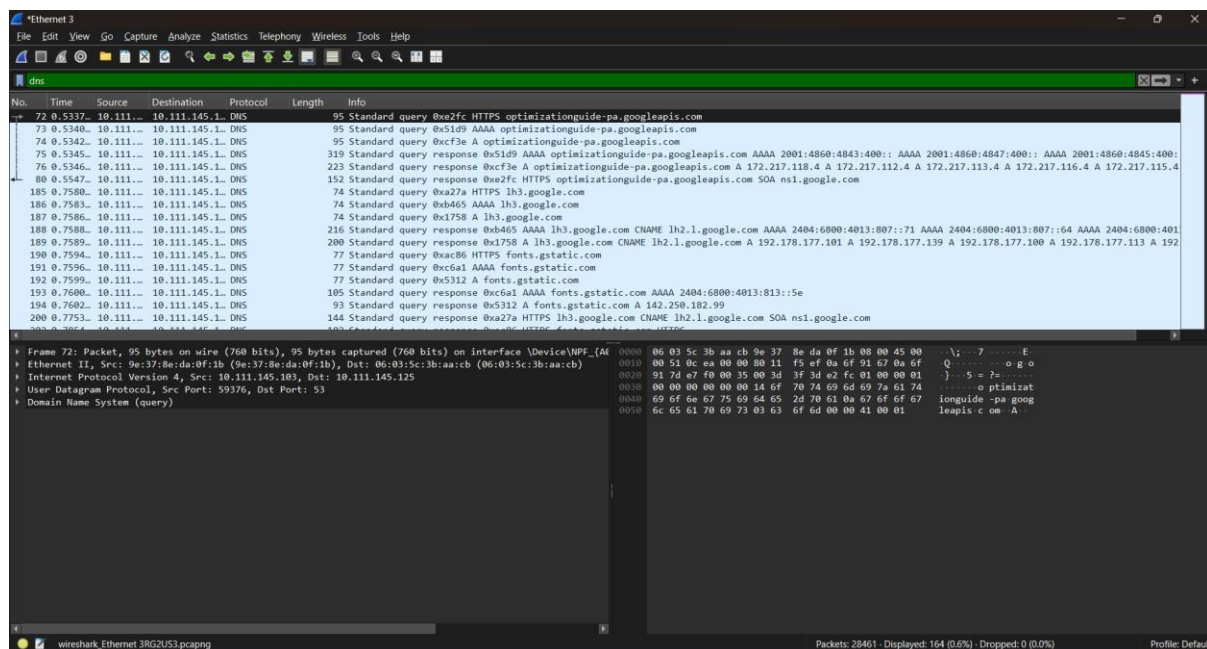


Day 2: TCP/IP Suite Theory :What is the TCP/IP Suite? Layers of TCP/IP: Application, Transport, Internet, and Network Access. Practical :Analyze packet headers to identify the TCP/IP layer using Wireshark. Task: Compare OSI and TCP/IP models with real-world examples.

Tasks:-

1. Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

Ans.:-



TCP/IP Layer packet		Header fields you can mention
Application	DNS	Transaction ID, Flags, Queries
Transport	UDP	Source Port: 63514, Destination Port: 53
Internet	IPv4	Source IP: 10.111.145.103, Destination IP: 10.111.145.125
Network		
Ethernet II		Source MAC, Destination MAC
Access		

2. Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

Ans.:-

OSI Model Layer	TCP/IP Model Layer	Real-World Example (WhatsApp / Zomato)
7. Application	Application	You type and send a WhatsApp message or place a Zomato order .
6. Presentation	Application	WhatsApp encrypts your message and converts it into a readable format.
5. Session	Application	WhatsApp keeps your chat session active while you exchange messages.
4. Transport	Transport	TCP ensures your WhatsApp message or Zomato order reaches the server correctly.

OSI Model Layer	TCP/IP Model Layer	Real-World Example (WhatsApp / Zomato)
3. Network	Internet	IP routes the message/order from your phone to the WhatsApp or Zomato server over the Internet.
2. Data Link	Network Access	Wi-Fi or Ethernet sends data frames between your phone/laptop and the Wi-Fi router.
1. Physical	Network Access	Radio signals (Wi-Fi) or Ethernet cable physically carry the data bits.

**3. Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.

Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.**

Ans.:-

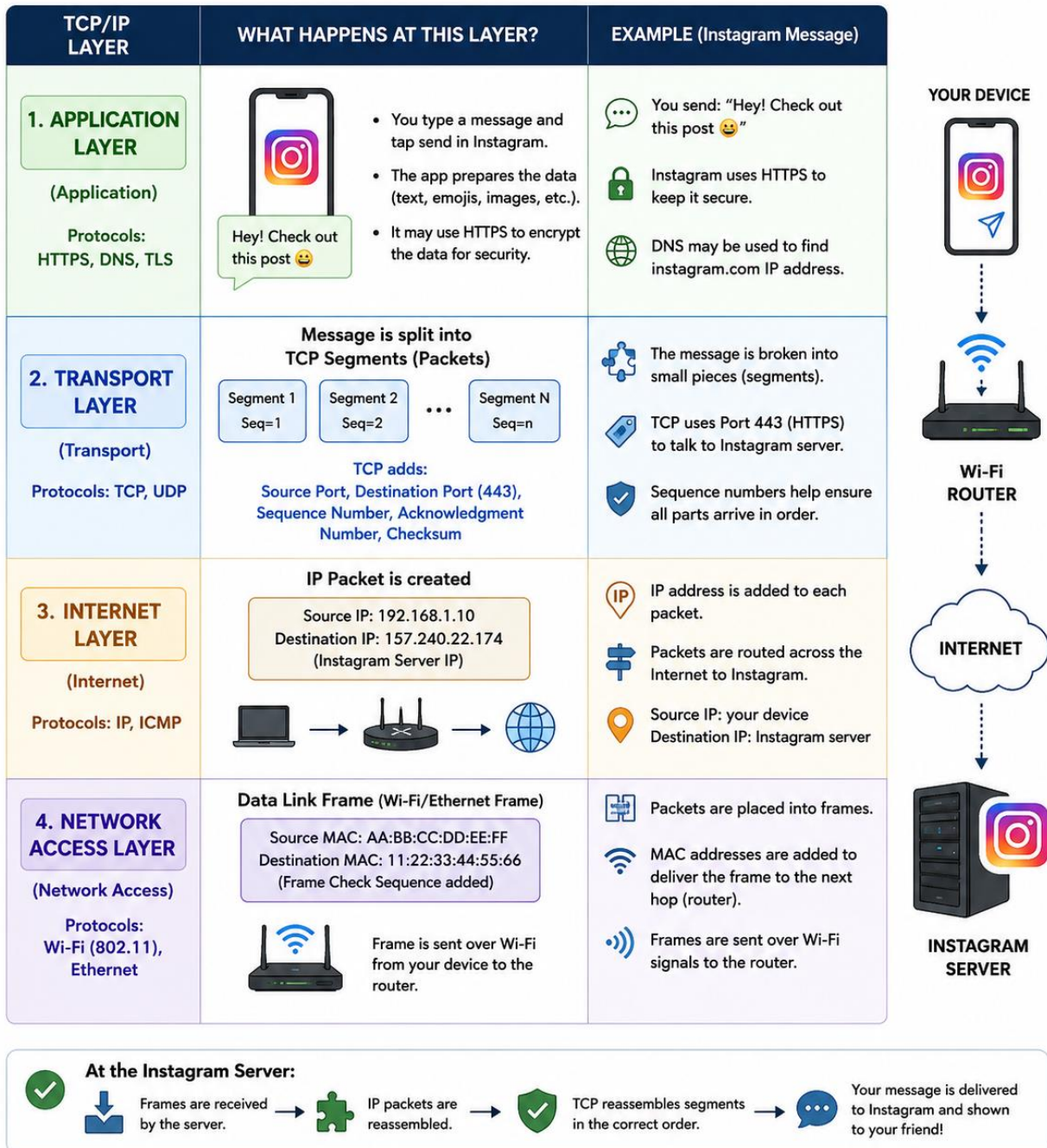
TCP Header Field	Transport Layer	Purpose
Source Port	Yes	Identifies the application or service that sent the data (e.g., browser).
Destination Port	Yes	Identifies the application or service that should receive the data (e.g., HTTPS uses port 443).
Sequence Number	Yes	Indicates the position of the first byte in the segment, helping keep data in the correct order.
Acknowledgment Number	Yes	Confirms receipt of data by indicating the next expected byte.

TCP Header Field	Transport Layer	Purpose
Header Length (Data Offset)	Yes	Specifies the size of the TCP header so the receiver knows where the data begins.
Flags (SYN, ACK, FIN, RST, PSH, URG)	Yes	Control connection setup, acknowledgment, termination, and data handling.
Window Size	Yes	Indicates how much data the receiver can accept before requiring another acknowledgment.
Checksum	Yes	Detects errors in the TCP header and data during transmission.
Urgent Pointer	Yes	Points to urgent data when the URG flag is set.
Options (if present)	Yes	Provides additional features such as Maximum Segment Size (MSS), Window Scaling, or Timestamps.

4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).

Ans. :-

How an Instagram Message Travels from Your Device to Instagram Server (Using TCP/IP Model)



5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.

Ans.:-

Selected Packet : TCP Packet (Destination Port 443 - HTTPS)

TCP/IP Layer: Transport Layer

How I Identified It:-

In Wireshark, the Protocol column showed TCP. The packet details included Source Port, Destination Port (443), Sequence Number, and Acknowledgment Number, which are TCP header fields.

Real-World App Action:-

This packet represents the reliable transfer of data when sending a WhatsApp message, logging into Instagram, or booking a movie ticket on BookMyShow. TCP ensures that all packets are delivered correctly and in the correct order before the app processes the request.